

Lab 02 – First Contact with OpenPLC

Industrial Security (Bachelor)

• Maps to Section 02

• 30–45 min

LAB 02

🎯 Goal

Bring up the lab Docker stack and confirm that the soft PLC accepts and runs a ladder-logic program. End the lab with a Modbus read working against it.

✂ Step 1 – Bring up the stack

```
1 | cd bachelor/labs
2 | docker compose up -d
3 | docker compose ps           # all services should be (healthy)
```

✂ Step 2 – Open the OpenPLC web UI

Browse to <http://127.0.0.1:18080>. Log in with the default credentials shown by the Open-PLC project; change the password right away.

✂ Step 3 – Upload the ladder program

Upload the file `scripts/ladder/conveyor.st`. Click **Start PLC**. Open the **Monitoring** page and watch **Motor** toggle when you flip **Start**.

✂ Step 4 – Read holding registers

```
1 | docker compose exec student python3 /labs/scripts/modbus_read.py
```

You should see ten register values. They will all be zero unless the program has written to them.

📁 Deliverable

Hand in: (a) a screenshot of the OpenPLC *Monitoring* page with your running program, and (b) the terminal output of `modbus_read.py`.

⚠ Caution

The Modbus listener is bound to `127.0.0.1:5502` on your host. Do not expose it to your network.